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Introduction

Football has become increasingly data-driven, with clubs, media organisations, betting companies and sports analytics firms relying on statistical analysis to support decision-making.

Predicting match outcomes remains challenging because results are influenced by multiple factors including team form, historical performance, home advantage and head-to-head records.

This project proposes the development of a machine learning classification system capable of predicting match outcomes across Europe's major football leagues. Using historical match data from the English Premier League, English Championship, Spanish La Liga, Italian Serie A, German Bundesliga and French Ligue 1, the project will identify patterns in football performance data and evaluate the effectiveness of predictive modelling techniques.

Problem Definition

Soccer in general is a field that is of great interest world-wide and accurate football forecasting has significant value for sports analytics organisations, newspapers, football clubs and betting sites.

Today, many football clubs have employed data experts to analyse in-play statistics to help improve team performance and identify player qualities or weakness. Sports betting companies are increasing their use of historical match results to help decide match odds, in an industry where high financial rewards are at stake.

The challenge is determining whether machine learning can improve prediction accuracy through the analysis of large-scale historical football datasets. Despite the availability of large volumes of football data, accurately predicting match outcomes remains a complex challenge due to the influence of numerous factors.

This project seeks to address this challenge by developing predictive classification models capable of forecasting match outcomes using objective historical performance data.

Why choose this field of study?

The Data Preparation and Feature Engineering aspects of this project will require knowledge of the field of study. As a follower of football, I have a good understanding of the important variables required and can quickly spot any data issues. I will also understand the Target Variables and which columns will be important to train for machine learning. This will also become an interesting learning curve into the machine learning process.

Aim and Objectives

Aim

The aim of this project is to develop and evaluate machine learning classification models capable of predicting football match outcomes using historical data from Europe's major football leagues.

The project will investigate whether predictive analytics techniques can improve forecasting accuracy by analysing historical match statistics, identifying influential performance factors, and comparing the effectiveness of multiple machine learning algorithms.

Objectives

1. Collect and integrate football match data from multiple European leagues.
2. Perform exploratory data analysis to identify influential factors affecting match outcomes.
3. Engineer predictive features such as team form and home advantage.
4. Develop and compare machine learning classification models.
5. Evaluate model performance using industry-standard metrics.
6. Present findings through reporting and dashboard visualisation.

Business Objective

To determine whether machine learning models can predict football match outcomes more accurately than simple rule-based approaches by analysing historical football data.

Scope

Included:

Data from Top European Football Leagues, such as:

England	Premier League
England	Championship
Germany	Bundisliga
Italy	Serie A
Spain	La Liga
France	Ligue 1

The selected leagues above, will provide sufficient data to support machine learning model development. The project is expected to generate a dataset of over 20,000 records, which should provide a foundation for identifying patterns and train predictive models to evaluate model performance.

Excluded:

Live prediction

Real time betting

In play forecasting

Individual player predictions

Deep learning approaches

Methodology

Phase 1

Data Collection:

The football match data from 2014/2015 to 2025/2026 will be taken from six major European leagues. This will provide a large volume of historical data, which will be suitable in identifying trends associated with football outcomes.

Phase 2

The source used for downloading csv files for the 6 European leagues will be:

https://football-data.co.uk/data.php

In total 72 csv files will be downloaded. (12 season x 6 leagues)

For each of the 12 seasons from every league, I will create one master csv file, for each of the six leagues mentioned.

This will be achieved by importing the csv files for each league and merging them using the excel data model. As the format and column heading remains the same for all csv files, this shouldn't involve much data cleaning, at this stage.

Phase 3

Data Preparation

The data preparation stage is critical for the overall quality of the project, as the machine learning models depend on the quality of the data received. The tasks here include:

- Renaming Columns
- Remove Columns not required
- Handling missing values
- Transforming Data to ensure suitability for modelling
- Creating derived columns for analysis

Phase 4

Integrated Football Dataset

- Merge 6 Master csv files that were imported, into 1 data frame, using Python

Phase 5

Feature Engineering

- Team Form
- Home Advantage
- Rolling Statistics

Feature engineering will play an important role in model performance. New variables will be created from the dataset to provide additional predictive analysis. These features will attempt to capture trends that may not be apparent with original variables alone.

Phase 6

Exploratory Data Analysis

- Goal Trends
- Win Percentage Analysis
- League comparisons

Phase 7

Machine Learning Models

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

Machine learning algorithms will be implemented and compared to determine which approach provides the greatest predictive accuracy. Logistic regression will be the baseline classification model. Decision Tree and Random Forest models will allow investigation of non-linear relationships. Comparing several approaches will help identify the model most suitable for the match outcome prediction.

Phase 8

Model Evaluation

Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Phase 9

Visualisation & Final Report

- Performance reporting
- Model comparison



Figure 1 Proposed project Methodology and Workflow

Data Sources

The primary data source used for this project will be:

https://football-data.co.uk/data.php

This is a publicly available repository containing the required historical match statistics.

The data held within the csv files include important match data, such as, goals scored, shots on target, home goals, away goals and match results.

These variables will be relevant to the proposed business problem as they provide the information required for machine learning models to identify patterns associated with predicting football match outcomes.

Expected Deliverables

The expected deliverables for this project, by the end of the second semester include:

Data Collection

Find and download the desired files for the six European leagues. This will require creating csv files for each of the 12 seasons, for the six leagues. Use the Excel Data Model to merge the 12 csv files into one file for each of the leagues.

Data Preparation

Using Python, I will run a data preparation process to ensure all six csv files have been reviewed for missing data, errors and creating any required derived variables.

Integrated Football Dataset

Creating an integrated football dataset by merging the six csv files mentioned above. This merged dataset will be cleaned and ready for analytical and predictive modelling purposes. Also ensure that the final merged dataset, has only required columns, deleting any redundant variables.

Exploratory Data Analysis

Run an analysis of the dataset to identify any trends and relationships across the leagues. Create visualisations and summaries to report any findings

Model Evaluation

Build a trained and evaluated machine learning classification model, capable of predicting football match outcomes.

Compare model performance, such as Accuracy, Precision, Recall and F1-Score metrics. Create a Confusion Matrix analysis to identify effective predictive model.

Final Capstone Report

Create a comprehensive report documenting the business problem, methodology, data preparation, modelling process, results and conclusions.

Project Timeline



Figure 2 Proposed Project Timeline 2026-2027

Ethical and Legal Considerations

Although the project uses publicly available football match data, the ethical and legal considerations remain an important issue.

The dataset contains only historical football match information and does not contain personal, confidential or sensitive data and therefore does not raise any GDPR or privacy concerns,

The data will be used solely for research and academic purposes in accordance with the website's own restrictions.

This project will also provide all steps taken in python for Data preparation, feature engineering and modelling decisions.

Any external sources and supporting materials will be appropriately referenced using Harvard Referencing standards.

The use of Artificial intelligence tools will be documented in accordance with the AI Fluency Framework, specified by CCT College

References

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Appendix

Appendix A – Reflection on the use of Artificial Intelligence

Artificial Intelligence (Microsoft 365 Copilot) was used throughout this project as a support tool to assist with report planning and guidance and refine the project scope.

During the early stages, Copilot was used to explore project topics and evaluate suitable datasets. In discussion with Copilot, the initial idea has evolved into a machine learning project aimed at predicting football matches

A major benefit of AI was the ability to provide examples on finding my dataset, without causing any GDPR issues.

AI was also used to review draft interpretations and suggest alternative wording for my report. These suggestions were reviewed critically and adapted where appropriate, before being incorporated into the final report. AI was used as a supporting tool and not as a substitute for independent learning.